Lecture 29 — Characteristic equation, $2imes2$ eigenproblems, and complex numbers

Wed Nov 18

Last updated: October 29, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §6.1 (PDF) — characteristic equation / complex example
Video: 18.06 Lec 21 — eigenvalues and eigenvectors · 18.06 Lec 22 — diagonalization and powers of A · MIT 18.06 playlist (Strang)

Reading: Strang §6.1, characteristic equation and the complex-eigenvalue (90° rotation) example. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

In the story. Monday: eigenvectors = persistent directions ($A^t v = \lambda^t v$). Today: **compute** them for 2×2 via $\det(A - \lambda I) = 0$, including the complex case when there is no real eigenpair. Friday: Lab 5 iterates the campus matrix and explains the long-run mix with $\lambda = 1$.

Learning objectives

By the end of class, you should be able to:

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors.
5. Check real or complex eigenpairs by substitution.

Warm-up

Monday's rotation $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ had no real eigenvector.

(a) Form $R - \lambda I$ and compute $\det(R - \lambda I)$ as a quadratic in λ.

(b) What are the roots? Are they real?

Fill in

Complex numbers. A complex number is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 =$ ____. Plot z as the point (a, b) in the **complex plane**.

Arithmetic (what we need):

- Add/subtract: combine real and imaginary parts.

- Multiply: $(a+bi)(c+di) = (ac-bd)+(ad+bc)i$.
- Conjugate: $\overline{a+bi} =$ _____.
- Magnitude: $|a+bi| = \sqrt{a^2+b^2}$.

Characteristic equation. Eigenvalues of square A solve

$$\det(A - \lambda I) = 0.$$

For 2×2, this is a _____ equation in λ.

Eigenvectors. For each eigenvalue λ, solve

$$(A - \lambda I)v = 0, \qquad v \neq 0$$

(null space of $A - \lambda I$). Scale freely — any nonzero multiple is fine.

Check. Always verify $Av = \lambda v$ by substitution (works for real or complex pairs).

Worked in class

W1. Complex drill. Let $z = 2 + i$ and $w = 1 - 3i$.

(a) Compute $z + w$, zw, and $\overline{z}$.
(b) Plot z and $\overline{z}$ in the complex plane.

W2. Real 2×2 by hand. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

(a) Expand $\det(A - \lambda I)$ and solve for λ.
(b) For each eigenvalue, find one eigenvector.
(c) Check one pair by computing Av and λv.

W3. Rotation eigenvalues. For $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$:

(a) Confirm the characteristic polynomial is $\lambda^2 + 1 = 0$, so $\lambda = \pm i$.
(b) For $\lambda = i$, solve $(R - iI)v = 0$ and give one complex eigenvector.
(c) Check $Rv = iv$ by matrix–vector multiplication (complex arithmetic).

W4. Interpretation preview. Complex conjugate eigenvalues of a real matrix often signal _____ (rotation / oscillation) in the real plane — Friday's lab makes this concrete with software.

Your turn

Y1. Compute $(3 - 2i)(1 + i)$ and $|3 - 2i|$.

Y2. For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, form $\det(A - \lambda I)$, find both eigenvalues, and an eigenvector for $\lambda = 3$.

Y3. Check: with $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $v = (1, -i)$, verify $Av = iv$ (or explain the correct λ if this v is for $\lambda = -i$).

Y4. True or false: every real 2×2 matrix has two real eigenvalues.

Answers (Your turn)

Y1. $(3-2i)(1+i) = 3+3i-2i-2i^2 = 5+i$. Magnitude: $|3-2i| = \sqrt{9+4} = \sqrt{13}$.

Y2. $A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{bmatrix}$, so $\det = (1-\lambda)(3-\lambda)$. Eigenvalues $\lambda = 1$ and $\lambda = 3$. For $\lambda = 3$: $(A-3I)v = \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} v = 0$ gives $v_1 = v_2$, e.g. $v = (1,1)$.

Y3. $Av = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix}$ and $iv = \begin{bmatrix} i \\ 1 \end{bmatrix}$. Yes — $Av = iv$ (so this v matches $\lambda = i$).

Y4. **False.** The characteristic polynomial may have complex roots (e.g. 90° rotation: $\lambda = \pm i$).